Stock Project: Stage 5 Report (ps5-sol2)

Team: Stock Project | Member: Scottie Dunnavan | CSCI 4425/5425 Introduction to Deep Learning

1. Problem Statement

Retail traders compete with institutions that trade on proprietary order-book data. This project studies short-horizon price-direction prediction for small-to-mid-cap U.S. equities. For each time step, models estimate the probability that the price moves Up (2), stays Neutral (1), or moves Down (0) within a spread-adjusted barrier. The production horizon is 180 seconds. The offline verification pipeline uses 50 ticks. Evaluation is chronological and leakage-aware. A well-evaluated negative result is a valid outcome.

Refinements since PS4:

- Universe: one market-cap band, 1M to 1B USD, computed once before the session. This replaces the 1M to 500M band and removes a dependency on a later-stage catalyst agent.
- No minimum price floor. Sub-1 USD equities are kept. Reg NMS Rule 612 allows sub-penny quoting below 1 USD, and the spread-adjusted barrier scales with proportional spread.

2. Objectives and Status

ID	Objective	Status
O1	Reproduce DeepLOB on FI-2010	Planned
O2	Majority-class and Random Forest baselines	Implemented (src/preprocessing.py); XGBoost planned
O3	LSTM-Attention and DeepLOB with auxiliary flow branch	Specified; not yet trained
O4	Macro-F1, per-class F1, confusion matrix	Implemented; calibration (ECE) and day-block intervals planned
O5	Live dashboard and paper trading	Stretch; not started

3. Dataset

Planned data (not yet collected):

- FI-2010 (validation): 5 Nasdaq Nordic stocks, 10 days, about 4 million events, top-10 levels. Used to reproduce DeepLOB and pretrain convolutional layers. Its labels use event-count horizons, so they are not directly comparable to our 180-second results.
- Databento historical (bridging): EQUS.MINI consolidated quotes and trades for labels and 1-minute macro bars. ARCX.PILLAR (NYSE Arca) and XNAS.ITCH (Nasdaq) provide 10-level

books. Cost is checked before each pull.

- Live recording (ongoing): tastytrade dxLink consolidated quotes and trades. Moomoo ArcaBook 10-level order book (order-book subscription only). IBKR TotalView top-3 depth (the EDS add-on is required for API use).

Current verification data: a synthetic order-book generator (Section 5). It is not market data.

Schemas (implemented in src/preprocessing.py):

- l1_quotes.parquet: 18 columns, including session, venue fields (bid_exch, ask_exch, trade_exch), Lee-Ready trade_side (+1, -1, 0), and cumulative volume.
- l2_snapshots.parquet: 47 columns, 10 levels of bid and ask price and size, plus session and book_age_ms.
- Writes are atomic: a temporary file, then an atomic rename.

4. Solution

Implemented in src/preprocessing.py. src/main.py runs the pipeline and prints the results table.

- Spatial features: relative price distance (p_i - p_mid) / p_mid and log size ln(1 + v) for 10 levels per side, giving 40 values per step. Stacked over 100 steps into a (Batch, 1, 100, 40) tensor.
- Order-flow imbalance (OFI), keyed by exact price across 10 levels. The best-level form is defined by Cont, Kukanov and Stoikov (2014). The multi-level form follows Xu, Gould and Howison (2019). Entry and exit guards keep levels that scroll in or out of the top 10 from being counted as flow. A level that vanishes without scrolling out is assigned to the nearest current level, which is an approximation (Section 6).
- Absorption and iceberg ratio (AIR) at the top of book: aggressive trade volume divided by top-level size depletion, floored at 1. Venue separation is not enforced in code. The synthetic stream is single-venue, so the venue-matched definition in the v1.2 spec is not yet implemented. The ratio is reported as continuous. Threshold flags are not implemented.
- Triple-barrier labels (Lopez de Prado, 2018): Up if the future best bid clears the current best ask by the barrier first; Down if the future best ask falls below the current best bid by the barrier first; otherwise Neutral. The barrier scales with trailing volatility, with a 15 bps minimum.
- Trade direction: Lee and Ready (1991) quote rule with a tick-test fallback.
- Chronological 70/30 split. The training set drops the last 50 samples before the split, which equals the label horizon, so training labels do not reach into the test window.
- Models, all trained in src/preprocessing.py:
    - Majority class.
    - Random Forest (150 trees, balanced class weights).
    - A shallow multilayer perceptron (hidden layers of 128 and 64 units) on standardized features. This is a neural baseline. It is not a deep model and not a late-fusion architecture.

Tensor shapes produced: DeepLOB spatial (N, 1, 100, 40); auxiliary flow (N, 100, 16); tabular (N, 56).

Planned for the production run on NMSU simurgh (not implemented yet):

- 600-tick window (60 s) and 180-second horizon. Volatility from 15-second NBBO mid returns over a 15-minute window, scaled by sqrt(12).
- PyTorch LSTM-Attention (2-layer LSTM, hidden size 128, temporal self-attention) on 60 one-minute bars.
- DeepLOB with an auxiliary flow branch: 2-epoch frozen convolution warmup, then 0.1x learning rate.
- Temperature scaling (ECE) and day-block bootstrap intervals.

5. Preliminary Results

Source: `python src/main.py` on the synthetic generator.

Samples: 850 labelled (Down 348, Neutral 140, Up 362). Chronological 70/30 split with a 50-sample purge. Test n = 255.

Shapes: DeepLOB spatial (850, 1, 100, 40); aux flow (850, 100, 16); tabular (850, 56).

```
Model                    Macro-F1   F1 Down   F1 Neut   F1 Up
Majority Class           0.1696     0.0000    0.0000    0.5088
Random Forest            0.4192     0.6920    0.0370    0.5285
MLP (scikit-learn)       0.4549     0.6463    0.2532    0.4653
```

Random Forest confusion matrix (rows = true [Down, Neutral, Up], cols = predicted):

```
[91,  5, 34]
[16,  1, 21]
[26, 10, 51]
```

MLP confusion matrix (same row order):

```
[74, 11, 45]
[ 5, 10, 23]
[20, 20, 47]
```

Interpretation:

- Both models beat the majority-class baseline (0.170 Macro-F1).
- Neutral is the hardest class. Random Forest recovers 1 of 38 Neutral samples; the MLP recovers 10.
- The MLP leads on Macro-F1 (0.455 vs 0.419). With 255 test samples and no confidence intervals, this gap is not established as significant.

Important: the generator links order-book imbalance and trade side to the direction of the next price move. Scores above chance therefore reflect how the data was constructed, not market predictability.

These results verify that the pipeline runs end to end.

6. Limitations and Next Steps

- The only neural model in this submission is a shallow MLP. The DeepLOB and LSTM-Attention models, which form the deep-learning component of this project, are specified but not yet trained.
- All results come from synthetic data (see Section 5).
- The triple-barrier multiplier is k=0.45 in code. The locked spec sets an initial k=1.0, fit on the train split. Align in V2 and report the resulting class shares.
- OFI: a level that vanishes without scrolling out is assigned to the nearest current level. This is a known approximation, planned for v1.3.
- AIR: venue-matched definition and threshold flags (ICEBERG_ABSORPTION, UNEXECUTED_DEPLETION) are not implemented.
- The recorder's 5-minute buffered flush and bounded queue are planned, not implemented.
- FI-2010 reproduction has not been run.
- Next: pull Databento data after cost checks, run the causal Tier 2 replay, report non-overlapping sample counts, then train the deep models on simurgh.

7. References

[1] Zhang, Z., Zohren, S., & Roberts, S. (2019). DeepLOB: Deep convolutional neural networks for limit order books. IEEE Transactions on Signal Processing, 67(11), 3001-3012. [2] Ntakaris, A., Magris, M., Kanniainen, J., Gabbouj, M., & Iosifidis, A. (2018). Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods. Journal of Forecasting. [3] Cont, R., Kukanov, A., & Stoikov, S. (2014). The price impact of order book events. Journal of Financial Econometrics, 12(1), 47-88. [4] Xu, H., Gould, M. D., & Howison, S. D. (2019). Multi-level order-flow imbalance in a limit order book. Market Microstructure and Liquidity, 4(3-4). [5] Lee, C. M. C., & Ready, M. J. (1991). Inferring trade direction from intraday data. Journal of Finance, 46(2), 733-746. [6] Lopez de Prado, M. (2018). Advances in Financial Machine Learning. Wiley.